

Project name: PetConnect

1. Project Overview and Objectives

PetConnect is a proposed centralized web platform that will help pet owners and service providers in Salzburg manage pet care efficiently. It replaces fragmented tools such as WhatsApp groups and paper logs with a unified digital solution.

Problem Statement:

- Missed vaccinations and appointments
- No centralized system for pet care management
- Difficulty tracking pet health and activity

2. Target Group

Pet Owners:

- People who may need assistance with pet care (e.g., due to time constraints, lack of experience, or physical limitations)
- Located in Salzburg

Pain Points of Pet Owners:

- Forget appointments due to disorganized scheduling
- Use multiple tools for pet management
- Lack of overview of pet health data

Service Providers:

- Dog walkers
- Pet sitters

Pain Points of Service Providers:

- Disorganized scheduling
- Communication spread across platforms
- No centralized booking system

3. Personas

Persona 1 – Pet Owner

Name: Anna Müller

Age: 22

Occupation: Student / Part-time worker

Location: Salzburg

Goals:

- Keep her pet safe
- Manage appointments easily

Frustrations:

- Forgetting vaccinations
- Using multiple apps/tools like job boards, different WhatsApp groups

Persona 2 – Service Provider

Name: Lukas Steiner

Age: 35

Occupation: Dog Walker

Goals:

- Manage bookings efficiently
- Communicate with clients easily

Frustrations:

- Messy scheduling via WhatsApp/calls
- Lack of structured system

4. Usage Context / Environment

Usage Scenarios:

- On the go (mobile devices)
- At home (checking pet data)
- At work (managing bookings)

Requirements:

- Responsive (mobile-friendly) design
- Simple and intuitive navigation

5. Functional Requirements

User Management

- Registration and login
- Role-based access (Owner / Provider)

Digital Pet Passport

- Store pet details (name, breed, age)
- Vaccination records

Health Dashboard

- Track weight
- Visual representation (charts)

Booking & Scheduling

- Book services
- View availability
- Cancel/Confirm appointments

Messaging

- Direct communication between users

6. Data Sources

- **Primary Sources:** pet datasets from Kaggle and manually populated data by the project team. External datasets will provide breed, age, weight, vaccination, and pet image information, while manually populated data will complete missing attributes.
- **Data Quality Considerations:** external datasets and manually entered data may contain inaccuracies; therefore, validation (e.g., weight ranges, age limits) and data cleaning will be applied before integration.

7. Technical Concept

Frontend (Presentation Layer)

- ReactJS
- Responsive UI design

Backend (Logic Layer)

- NestJS (REST API)
- Authentication (JWT)

Database (Data Layer)

- PostgreSQL

Data Format

- JSON

8. Design & Usability

- Clean and minimal interface
- Dashboard overview for quick insights
- Easy booking process
- Clear navigation structure

9. Task Distribution

- **Data Lead:** Yeabsira Tekuamwork Bizuneh
 - Responsible for data sourcing, data handling, database schema design, and data preprocessing in collaboration with Abirha Kahsay.
- **Programming Lead:** Abirha Kahsay Berhe
 - Responsible for backend and frontend implementation in collaboration with Yeabsira Tekuamwork.
- **UI/Viz Lead:** Sadii Makavita Gamage
 - Responsible for dashboard design, user interface, usability, and frontend development.